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PUMP UNIT AND AIR SUPPLY DEVICE

Abstract

A pump unit includes a motor including a rotation shaft configured to rotate about a rotational axis, a first pump and a second pump, a first drive transmission portion configured to be transitioned between a first transmission state and a first non-transmission state, and a second drive transmission portion configured to be transitioned between a second transmission state and a second non-transmission state. The first drive transmission portion is configured to drive the first pump by being transitioned to the first transmission state, and configured not to drive the first pump by being transitioned to the first non-transmission state. The second drive transmission portion is configured not to drive the second pump by being transitioned to the second non-transmission state, and configured to drive the second pump by being transitioned to the second transmission state.

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Background/Summary

BACKGROUND OF THE INVENTION
Field of the Invention
[0001] The present invention relates to a pump unit and an air supply device for conveying air using a pump.
Description of the Related Art
[0002] Hitherto, an image forming apparatus in which is installed a cartridge equipped with a developing container that accommodates toner and an attachment portion capable of having a supply pack accommodating toner attached thereto is proposed (refer to Japanese Patent Application Laid-Open Publication No. 2023-134401). Toner discharged from the supply pack attached to the attachment portion is accommodated temporarily in a reserve tank disposed on a cartridge, and conveyed from the reserve tank to the developing container. Toner is conveyed from the reserve tank to the developing container by a screw or air flowing by a pump.
SUMMARY OF THE INVENTION
[0003] According to a first aspect of the present invention, a pump unit includes a motor including a rotation shaft configured to rotate about a rotational axis in a first rotation direction and in a second rotation direction opposite to the first rotation direction, the rotation shaft including a first end portion on one side of the rotation shaft and a second end portion on the other side of the rotation shaft in a direction of the rotational axis, a first pump and a second pump that are configured to send out air, a first drive transmission portion configured to be transitioned between a first transmission state in which a first driving force is transmitted from the first end portion of the rotation shaft to the first pump and a first non-transmission state in which the first driving force is not transmitted from the first end portion to the first pump, and a second drive transmission portion configured to be transitioned between a second transmission state in which a second driving force is transmitted from the second end portion of the rotation shaft to the second pump and a second non-transmission state in which the second driving force is not transmitted from the second end portion to the second pump. The first drive transmission portion is configured to drive the first pump by being transitioned to the first transmission state when the rotation shaft rotates in the first rotation direction, and configured not to drive the first pump by being transitioned to the first non-transmission state when the rotation shaft rotates in the second rotation direction. The second drive transmission portion is configured not to drive the second pump by being transitioned to the second non-transmission state when the rotation shaft rotates in the first rotation direction, and configured to drive the second pump by being transitioned to the second transmission state when the rotation shaft rotates in the second rotation direction.
[0004] According to a second aspect of the present invention, an air supply device for conveying toner in an image forming apparatus includes a motor including (i) a housing, and (ii) first and second shaft portions which are rotatable together about a rotational axis in a first rotation direction and a second rotation direction opposite to the first rotation direction, and which extend in opposite directions with each other with respect to the housing in a direction of the rotational axis, first and second air supply unit configured to supply air, a first clutch configured to transmit a first driving force from the first shaft portion to the first air supply unit when the motor is rotated in the first rotation direction, and configured not to transmit the first driving force from the first shaft portion to the first air supply unit when the motor is rotated in the second rotation direction, and a second clutch configured to transmit a second driving force from the second shaft portion to the second air supply unit when the motor is rotated in the second rotation direction, and configured not to transmit the second driving force from the second shaft portion to the second air supply unit when the motor is rotated in the first rotation direction.

[0005] According to a third aspect of the present invention, an air supply device for conveying toner in an image forming apparatus includes a motor including (i) a housing, and (ii) first and second shaft portions which are rotatable together about a rotational axis in a first rotation direction and a second rotation direction opposite to the first rotation direction, and which extend in opposite directions with each other with respect to the housing in a direction of the rotational axis, first and second air supply unit configured to supply air, a first clutch configured to be transitioned between a first transmission state in which a first driving force is transmitted from the first shaft portion to the first air supply unit, and a first non-transmission state in which the first driving force is not transmitted from the first shaft portion to the first air supply unit, and a second clutch configured to be transitioned between a second transmission state in which a second driving force is transmitted from the second shaft portion to the second air supply unit, and a second non-transmission state in which the second driving force is not transmitted from the second shaft portion to the second air supply unit.
[0006] Further features of the present invention will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0007] FIG. **1** is a perspective view of a pump unit according to the present embodiment.
[0008] FIG. **2**A is a plan view of the pump unit.
[0009] FIG. **2**B is a plan view of the pump unit having a first pipe, a second pipe, a third pipe, and a fourth pipe described below removed therefrom.
[0010] FIG. **2**C is a front view of the pump unit.
[0011] FIG. **3** is an exploded perspective view of the pump unit.
[0012] FIG. **4**A is a plan view of a fourth clutch portion in a transmission state.
[0013] FIG. **4**B is a plan view of the fourth clutch portion in a non-transmission state.
[0014] FIG. **5**A is a cross-sectional view of a **5**A-**5**A cross-section of FIG. **2**.
[0015] FIG. **5**B is a cross-sectional view of a first pump illustrating a state in which a first eccentric shaft has been rotated for 90 degrees from FIG. **5**A.
[0016] FIG. **5**C is a cross-sectional view of the first pump illustrating a state in which the first eccentric shaft has been rotated for 90 degrees from FIG. **5**B.
[0017] FIG. **5**D is a cross-sectional view of the first pump illustrating a state in which the first eccentric shaft has been rotated for 90 degrees from FIG. **5**C.
[0018] FIG. **6**A is a cross-sectional view of the first pipe.
[0019] FIG. **6**B is a perspective view of the first pipe.
DESCRIPTION OF THE EMBODIMENTS
[0020] A pump unit **1** according to the present embodiment is disposed, for example, in an image forming apparatus for forming an image on a recording material. The image forming apparatus refers to an apparatus for forming an image on a sheet used as a recording medium based on an image information entered from an external PC or an image information read from a document, and which includes a printer, a copying machine, a facsimile, and a multifunction machine. Further, the image forming apparatus may have an auxiliary device such as an optional feeder, an image reading apparatus, or a sheet processing apparatus additionally connected to a main body having an image forming function, and the entire system having such an auxiliary device connected thereto also serves as one kind of an image forming apparatus.
[0021] The image forming apparatus includes an image forming unit for forming an image on a recording material, and the image forming unit includes a process cartridge having a developing container for accommodating toner, and a toner cartridge that accommodates toner to be supplied to

the developing container. Four process cartridges and four toner cartridges are provided, corresponding to each of the four colors of toner, which are yellow, magenta, cyan, and black. The pump unit **1** according to the present embodiment may convey air selectively to the four toner cartridges, and toner and air are conveyed from the toner cartridge to which air has been sent from the pump unit **1** toward the developing container.

[0022] FIG. **1** is a perspective view of the pump unit **1** according to the present embodiment. FIG. **2**A is a plan view of the pump unit **1**. FIG. **2**B is a plan view of the pump unit **1** having a first pipe **6**L**1**, a second pipe **6**L**2**, a third pipe **6**R**1**, and a fourth pipe **6**R**2** described below removed therefrom. FIG. **2**C is a front view of the pump unit **1**. FIG. **3** is an exploded perspective view of the pump unit **1**.

[0023] In the following description, directions X, Y, and Z are defined as illustrated in the drawings. As illustrated in FIG. **1**, a short direction of the pump unit **1** is denoted by an X axis, and a direction from a rear side toward a front side of the pump unit **1** is referred to as an X direction. The X direction may also be referred to as a second direction, a forward direction, or a front direction. Further, a downstream side in the X direction of the pump unit **1** may be referred to as a front side, and an upstream side thereof may be referred to as a rear side.

[0024] A rotation shaft direction of a first dual shaft motor **4**L and a second dual shaft motor **4**R of the pump unit **1** is denoted by a Y axis, and a direction from the first dual shaft motor **4**L toward the second dual shaft motor **4**R is referred to as a Y direction. The Y direction may also be referred to as a direction of a rotational axis AL**1**, a first direction, an axial direction, or a longitudinal direction. Further, a downstream side in the Y direction of the pump unit **1** may be referred to as a right side, and the upstream side thereof may be referred to as a left side.

[0025] An up-down direction of the pump unit **1** is denoted by a Z axis, and a direction from a lower side toward an upper side of the pump unit **1** is referred to as a Z direction. The Z direction may also be referred to as a third direction, an upper direction, a height direction, and a vertical direction. Further, a downstream side in the Z direction of the pump unit **1** may be referred to as an upper side, an upper surface side, and a top surface side, whereas an upstream side thereof may be referred to as a lower side, a lower surface side, and a bottom surface side.

[0026] The X axis, the Y axis, and the Z axis are mutually perpendicular to each other. For example, the X axis is perpendicular to the Y axis, and also perpendicular to the Z axis. A plane perpendicular to the X axis may be referred to as a YZ plane, a plane perpendicular to the Y axis may be referred to as a ZX plane, and a plane perpendicular to the Z axis may be referred to as an XY plane. For example, the XY plane is a horizontal plane. The X direction and the Y direction are directions along the horizontal XY plane, that is, the horizontal direction.

[0027] The pump unit **1** includes, as illustrated in FIG. **1**, the first dual shaft motor **4**L, the second dual shaft motor **4**R, a first drive transmission portion **5**L**1**, a second drive transmission portion **5**L**2**, a third drive transmission portion **5**R**1**, and a fourth drive transmission portion **5**R**2**. The pump unit **1** includes a first pump **3**L**1**, a second pump **3**L**2**, a third pump **3**R**1**, a fourth pump **3**R**2**, a first holding member **7**L, a second holding member **7**R, a first pipe **6**L**1**, a second pipe **6**L**2**, a third pipe **6**R**1**, and a fourth pipe **6**R**2**.

[0028] The first pump **3**L**1** conveys, or sends out, air to the first pipe **6**L**1** by having the drive of the first dual shaft motor **4**L transmitted via the first drive transmission portion **5**L**1**. The air sent to the first pipe **6**L**1** is conveyed, for example, to a toner cartridge accommodating yellow toner attached to the image forming apparatus.

[0029] The second pump **3**L**2** conveys air to the second pipe **6**L**2** by having the drive of the first dual shaft motor **4**L transmitted via the second drive transmission portion **5**L**2**. The air sent to the second pipe **6**L**2** is conveyed, for example, to a toner cartridge accommodating magenta toner attached to the image forming apparatus.

[0030] The third pump **3**R**1** conveys air to the third pipe **6**R**1** by having the drive of the second dual shaft motor **4**R transmitted via the third drive transmission portion **5**R**1**. The air sent to the third pipe **6**R**1** is conveyed, for example, to a toner cartridge accommodating cyan toner attached to the image forming apparatus.

[0031] The fourth pump **3**R**2** conveys air to the fourth pipe **6**R**2** by having the drive of the second dual shaft motor **4**R transmitted via the fourth drive transmission portion **5**R**2**. The air sent to the fourth pipe **6**R**2** is conveyed, for example, to a toner cartridge accommodating black toner attached to the image forming apparatus.

[0032] As illustrated in FIGS. **1** to **3**, the pump unit **1** serving as an air supply device includes an attachment member **37**, and the first dual shaft motor **4**L serving as a dual shaft motor is held by the first holding member **7**L attached to the attachment member **37**. The second dual shaft motor **4**R is held by the second holding member **7**R attached to the attachment member **37**.

[0033] The first dual shaft motor **4**L serving as a first motor and a motor includes, as illustrated in FIG. **3**, a first rotation shaft **4**Ls serving as a rotation shaft that extends in the Y direction, and a first housing **4**Lc serving as a housing that rotatably supports the first rotation shaft **4**Ls. The first rotation shaft **4**Ls passes through the first casing **4**Lc, and is configured rotatably about a rotational axis AL**1** serving as a first rotational axis and a second rotational axis in a first rotation direction R**1** and a second rotation direction R**2** opposite to the first rotation direction R**1**. Further, the first rotation shaft **4**Ls includes a first projection portion **4**La that projects from one side of the first casing **4**Lc in the Y direction, and a second projection portion **4**Lb that projects from the other side of the first casing **4**Lc. The first projection portion **4**La serving as a first end portion and a first shaft portion is disposed on one side of the first rotation shaft **4**Ls in the Y direction, and a second projection portion **4**Lb serving as a second end portion and a second shaft portion is disposed on the other side of the first rotation shaft **4**Ls in the Y direction. That is, the first projection portion **4**La and the second projection portion **4**Lb extend mutually in opposite sides in the Y direction with respect to the first housing **4**Lc. The first housing **4**Lc, the first pump **3**L**1**, and the second pump **3**L**2** are arranged in an aligned manner in the Y direction.

[0034] The second dual shaft motor **4**R serving as a second motor includes a second rotation shaft **4**Rs that extends in the Y direction, and a second housing **4**Rc that rotatably supports the second rotation shaft **4**Rs. The second rotation shaft **4**Rs passes through the second housing **4**Rc, and is configured rotatably about the rotational axis AL**1** in a third rotation direction R**3** and a fourth rotation direction R**4** opposite to the third rotation direction R**3**. Further, the second rotation shaft **4**Rs includes a third projection portion **4**Ra that protrudes from one side of the second housing **4**Rc in the Y direction, and a fourth projection portion **4**Rb that protrudes from the other side of the second housing **4**Rc. The third projection portion **4**Ra serving as a third end portion and a third shaft portion is disposed on one side of the second rotation shaft **4**Rs in the Y direction, and the fourth projection portion **4**Rb serving as a fourth end portion and a fourth shaft portion is disposed on the other side of the second rotation shaft **4**Rs in the Y direction. That is, the third projection portion **4**Ra and the fourth projection portion **4**Rb extend mutually in opposite sides in the Y direction with respect to the second housing **4**Rc. The second housing **4**Rc, the third pump **3**R**1**, and the fourth pump **3**R**2** are arranged in an aligned manner in the Y direction.

[0035] The first drive transmission portion **5**L**1** and the second drive transmission portion **5**L**2** are respectively attached to the first projection portion **4**La and the second projection portion **4**Lb of the first dual shaft motor **4**L. The first drive transmission portion **5**L**1** serving as a first clutch includes, as illustrated in FIGS. **2**A and **3**, a first clutch portion **5***d*L**1**, and a first eccentric shaft **5***e*L**1** disposed on an output member **5***c* of the first clutch portion **5***d*L**1**. The first clutch portion **5***d*L**1** includes an input member **5***a* to which drive from the first projection portion **4**La is entered, the output member **5***c*, and a clutch member **5***b* disposed between the input member **5***a* and the output member **5***c*. The clutch member **5***b* is driven to rotate following the rotation of the input member **5***a* that rotates together with the first projection portion **4**La. The clutch member **5***b* transmits drive to or shuts off the same from the output member **5***c*, as described below.

[0036] The first eccentric shaft **5***e*L**1** disposed on the output member **5***c* is provided eccentrically with respect to the first rotation shaft **4**Ls, and rotates integrally with the output member **5***c* that rotates about the first rotation shaft **4**Ls. The first eccentric shaft **5***e*L**1** is connected to the first pump **3**L**1** serving as a first air supply unit, and by the first eccentric shaft **5***e*L**1** rotating in the first rotation direction R**1** about the first rotation shaft **4**Ls, the first pump **3**L**1** operates. By the operation of the first pump **3**L**1**, air is discharged through a first discharge portion **35**L**1** of the first pump **3**L**1**. The first pipe **6**L**1** is connected to the first discharge portion **35**L**1**, as illustrated in FIGS. **1** and **2**A, and air discharged from the first discharge portion **35**L**1** is passed through the first pipe **6**L**1** and sent to a desired conveyance destination.

[0037] Similarly, the second drive transmission portion **5**L**2** serving as a second clutch includes, as illustrated in FIGS. **2**A and **3**, a second clutch portion **5***d*L**2**, and a second eccentric shaft **5***e*L**2** disposed on the output member **5***c* of the second clutch portion **5***d*L**2**. The second clutch portion **5***d*L**2** has a similar configuration as the first clutch portion **5***d*L**1**, and it is arranged symmetrically in the Y direction with respect to the first housing **4**Lc from the first clutch portion **5***d*L**1**.

[0038] The second eccentric shaft **5***e*L**2** disposed on the output member **5***c* of the second clutch portion **5***d*L**2** is provided eccentrically with respect to the first rotation shaft **4**Ls, and rotates integrally with the output member **5***c* that rotates about the first rotation shaft **4**Ls. The second eccentric shaft **5***e*L**2** is connected to the second pump **3**L**2** serving as a second air supply unit, and by the second eccentric shaft **5***e*L**2** rotating in the second rotation direction R**2** about the first rotation shaft **4**Ls, the second pump **3**L**2** operates. By the operation of the second pump **3**L**2**, air is discharged through a second discharge portion **35**L**2** of the second pump **3**L**2**. The second pipe **6**L**2** is connected to the second discharge portion **35**L**2**, as illustrated in FIGS. **1** and **2**A, and air discharged through the second discharge portion **35**L**2** is passed through the second pipe **6**L**2** and sent to a desired conveyance destination.

[0039] The third drive transmission portion **5**R**1** and the fourth drive transmission portion **5**R**2** are respectively attached to the third projection portion **4**Ra and the fourth projection portion **4**Rb of the second dual shaft motor **4**R. The third drive transmission portion **5**R**1** serving as a third clutch includes, as illustrated in FIGS. **2**A and **3**, a third clutch portion **5***d*R**1**, and a third eccentric shaft **5***e*R**1** disposed on the output member **5***c* of the third clutch portion **5***d*R**1**. The third clutch portion **5***d*R**1** adopts a similar configuration as the first clutch portion **5***d*L**1**.

[0040] The third eccentric shaft **5***e*R**1** disposed on the output member **5***c* of the third clutch portion **5***d*R**1** is provided eccentrically with respect to the second rotation shaft **4**Rs, and rotates integrally with the output member **5***c* that rotates about the second rotation shaft **4**Rs. The third eccentric shaft **5***e*R**1** is connected to the third pump **3**R**1** serving as a third air supply portion, and by the third eccentric shaft **5***e*R**1** rotating in the third rotation direction R**3** about the second rotation shaft **4**Rs, the third pump **3**R**1** operates. By the operation of the third pump **3**R**1**, air is discharged through a third discharge portion **35**R**1** of the third pump **3**R**1**. The third pipe **6**R**1** is connected to the third discharge portion **35**R**1**, as illustrated in FIGS. **1** and **2**A, and air discharged through the third discharge portion **35**R**1** is passed through the third pipe **6**R**1** and sent to a desired conveyance direction.

[0041] Similarly, the fourth drive transmission portion **5**R**2** serving as a fourth clutch includes, as illustrated in FIGS. **2**A and **3**, a fourth clutch portion **5***d*R**2**, and a fourth eccentric shaft **5***e*R**2** disposed on the output member **5***c* of the fourth clutch portion **5***d*R**2**. The fourth clutch portion **5***d*R**2** has a similar configuration as the first clutch portion **5***d*L**1**, and is arranged symmetrically in the Y direction with respect to the second casing **4**Rc from the third clutch portion **5***d*R**1**.

[0042] The fourth eccentric shaft **5***e*R**2** disposed on the output member **5***c* of the fourth clutch portion **5***d*R**2** is provided eccentrically with respect to the second rotation shaft **4**Rs, and rotates integrally with the output member **5***c* that rotates about the second rotation shaft **4**Rs. The fourth eccentric shaft **5***e*R**2** is connected to the fourth pump **3**R**2** serving as a fourth air supply portion, and by the fourth eccentric shaft **5***e*R**2** rotating in the fourth rotation direction R**4** about the second rotation shaft **4**Rs, the fourth pump **3**R**2** operates. By the operation of the fourth pump **3**R**2**, air is discharged through a fourth discharge portion **35**R**2** of the fourth pump **3**R**2**. The fourth pipe **6**R**2** is connected to the fourth discharge portion **35**R**2**, as illustrated in FIGS. **1** and **2**A, and air discharged through the fourth discharge portion **35**R**2** is passed through the fourth pipe **6**R**2** and sent to a desired conveyance destination.

Detailed Configuration of Clutch Portion

[0043] Next, detailed configurations of the respective clutch portions are described with reference to FIGS. **4**A and **4**B. The respective clutch portions have a similar configuration as described above, such that in the following description, the fourth clutch portion **5***d*R**2** is illustrated as an example. FIGS. **4**A and **4**B are each an enlarged view of the fourth clutch portion **5***d*R**2** of FIG. **2**A. FIG. **4**A illustrates the fourth clutch portion **5***d*R**2** in a transmission state, and FIG. **4**B illustrates the fourth clutch portion **5***d*R**2** in a non-transmission state. As described below, the fourth clutch portion **5***d*R**2** may be transited between the transmission state and the non-transmission state.

[0044] As illustrated in FIGS. **4**A and **4**B, the fourth clutch portion **5***d*R**2** includes the input member **5***a* into which drive from the fourth projection portion **4**Rb (refer to FIG. **3**) is entered, the output member **5***c*, and the clutch member **5***b* arranged between the input member **5***a* and the output member **5***c*. The clutch member **5***b* has a toothed surface **5***b***1** having a saw-tooth shape, and the output member **5***c* includes a toothed surface **5***c***1** that faces the toothed surface **5***b***1** and that is formed to have a saw-tooth shape.

[0045] As illustrated in FIGS. **3** and **4**A, when the second rotation shaft **4**Rs of the second dual shaft motor **4**R rotates in the fourth rotation direction R**4**, the input member **5***a* fixed to the fourth projection portion **4**Rb of the second rotation shaft **4**Rs also rotates in the fourth rotation direction R**4**. Then, the clutch member **5***b* also rotates in the fourth rotation direction R**4** following the rotation of the input member **5***a*. In this state, the input member **5***a* presses an inclined surface **5***b***2** of the clutch member **5***b*, such that the clutch member **5***b* moves in the Y direction, and the toothed surface **5***b***1** of the clutch member **5***b* is meshed with the toothed surface **5***c***1** of the output member **5***c*. Thereby, the fourth clutch portion **5***d*R**2** is in a transmission state in which the drive of the fourth projection portion **4**Rb is transmitted to the fourth pump **3**R**2**. That is, the drive of the fourth projection portion **4**Rb of the second rotation shaft **4**Rs is transmitted through the input member **5***a*, the clutch member **5***b*, and the output member **5***c* to the fourth eccentric shaft **5***e*R**2**, and the fourth pump **3**R**2** is operated by the rotation of the fourth eccentric shaft **5***e*R**2**.

[0046] Further, as illustrated in FIGS. **3** and **4**B, in a state where the second rotation shaft **4**Rs of the second dual shaft motor **4**R rotates in the third rotation direction R**3**, the input member **5***a* fixed to the fourth projection portion **4**Rb of the second rotation shaft **4**Rs also rotates in the third rotation direction R**3**. Then, the clutch member **5***b* also rotates in the third rotation direction R**3** following the rotation of the input member **5***a*. In this state, the input member **5***a* does not press the inclined surface **5***b***2** of the clutch member **5***b*, such that the clutch member **5***b* does not move in the Y direction, and the toothed surface **5***b***1** of the clutch member **5***b* remains separated from the toothed surface **5***c***1** of the output member **5***c*. Even if the toothed surfaces **5***b***1** and **5***c***1** are abutted against each other, due to their saw-tooth shapes, the toothed surfaces **5***b***1** and **5***c***1** will not mesh with each other.

[0047] Thereby, the fourth clutch portion **5***d*R**2** will be in a non-transmission state in which the drive of the fourth projection portion **4**Rb is not transmitted to the fourth pump **3**R**2**. That is, the rotation of the third rotation direction R**3** of the clutch member **5***b* will not be transmitted to the output member **5***c*, and the fourth pump **3**R**2** will not operate.

[0048] As described, according to the direction of rotation of the second rotation shaft **4**Rs of the second dual shaft motor **4**R, the fourth clutch portion **5***d*R**2** is switched between the transmission state and the non-transmission state, by which whether the fourth pump **3**R**2** is operated or not is determined. Such a configuration is also adopted in the first pump **3**L**1**, the second pump **3**L**2**, and the third pump **3**R**1**.

[0049] That is, as illustrated in FIG. **3**, when the first rotation shaft **4**Ls of the first dual shaft motor

4L rotates in the first rotation direction R**1**, the first clutch portion **5***d*L**1** will be in a transmission state and the second clutch portion **5***d*L**2** will be in a non-transmission state. In other words, the first drive transmission portion **5**L**1** will be in a transmission state in which drive force is transmitted from the first projection portion **4**La to the first pump **3**L**1**, and the second drive transmission portion **5**L**2** will be in a non-transmission state in which drive force is not transmitted from the second projection portion **4**Lb to the second pump **3**L**2**. Therefore, the first pump **3**L**1** will be driven by being transmitted a first driving force from the first projection portion **4**La of the first rotation shaft **4**Ls and the second pump **3**L**2** will not operate. Further, when the first rotation shaft **4**Ls of the first dual shaft motor **4**L rotates in the second rotation direction R**2**, the first clutch portion **5***d*L**1** will be in a non-transmission state and the second clutch portion **5***d*L**2** will be in a transmission state. In other words, the first drive transmission portion **5**L**1** will be in a non-transmission state in which drive force is not transmitted from the first projection portion **4**La to the first pump **3**L**1**, and the second drive transmission portion **5**L**2** will be in a transmission state in which drive force is transmitted from the second projection portion **4**Lb to the second pump **3**L**2**. Therefore, the first pump **3**L**1** will not operate and the second pump **3**L**2** will be driven by being transmitted a second driving force from the second projection portion **4**Ra of the first rotation shaft **4**Ls.

[0050] Further, when a second rotation shaft **4**Rs of a second dual shaft motor **4**R rotates in the third rotation direction R**3**, the third clutch portion **5***d*R**1** will be in a transmission state and the fourth clutch portion **5***d*R**2** will be in a non-transmission state. In other words, the third drive transmission portion **5**R**1** will be in a transmission state in which the drive force is transmitted from the third projection portion **4**Ra to the third pump **3**R**1**, and the fourth drive transmission portion **5**R**2** will be in a non-transmission state in which the drive force is not transmitted from the fourth projection portion **4**Rb to the fourth pump **3**R**2**. Therefore, the third pump **3**R**1** will be driven by being transmitted a third driving force from the third projection portion **4**Ra of the second rotation shaft **4**Rs and the fourth pump **3**R**2** will not be driven. Further, as described above, when the second rotation shaft **4**Rs of the second dual shaft motor **4**R rotates in the fourth rotation direction R**4**, the third clutch portion **5***d*R**1** will be in a non-transmission state and the fourth clutch portion **5***d*R**2** will be in a transmission state. In other words, the third drive transmission portion **5**R**1** will be in a non-transmission state in which the drive force is not transmitted from the third projection portion **4**Ra to the third pump **3**R**1**, and the fourth drive transmission portion **5**R**2** will be in a transmission state in which the drive force is transmitted from the fourth projection portion **4**Rb to the fourth pump **3**R**2**. Therefore, the third pump **3**R**1** will not be driven and the fourth pump **3**R**2** will be driven by being transmitted a fourth driving force from the fourth projection portion **4**Rb of the second rotation shaft **4**Rs.

[0051] Transmission states of the first clutch portion **5***d*L**1**, the second clutch portion **5***d*L**2**, the third clutch portion **5***d*R**1**, and the fourth clutch portion **5***d*R**2** may also be referred to respectively as a first transmission state, a second transmission state, a third transmission state, and a fourth transmission state. The states of the first drive transmission portion **5**L**1**, the second drive transmission portion **5**L**2**, the third drive transmission portion **5**R**1**, and the fourth drive transmission portion **5**R**2** at this time may also be referred to respectively as the first transmission state, the second transmission state, the third transmission state, and the fourth transmission state.

[0052] The non-transmission states of the first clutch portion **5***d*L**1**, the second clutch portion **5***d*L**2**, the third clutch portion **5***d*R**1**, and the fourth clutch portion **5***d*R**2** may also be referred to respectively as a first non-transmission state, a second non-transmission state, a third non-transmission state, and a fourth non-transmission state. Further, the states of the first drive transmission portion **5**L**1**, the second drive transmission portion **5**L**2**, the third drive transmission portion **5**R**1**, and the fourth drive transmission portion **5**R**2** at this time may also be referred to respectively as the first non-transmission state, the second non-transmission state, the third non-transmission state, and the fourth non-transmission state.

[0053] The configurations of the respective clutch portions described above are mere examples, and the present technique is not limited thereto. That is, the clutch portion may adopt other configurations as long as the cutoff or transmission of drive to the output shaft is changed. For example, other clutches such as an electromagnetic clutch that may be switched between a transmission state where drive force is transmitted and a non-transmission state where drive force is not transmitted regardless of the direction of rotation of the respective rotation shafts of the first dual shaft motor **4**L and the second dual shaft motor **4**R may be adopted instead of the respective clutch portions.

Detailed Configuration of Pump

[0054] Next, detailed configurations of the respective pumps will be described with reference to FIGS. **3** and **5**A to **5**D. The respective pumps adopt similar configurations as described above, such that in the following description, the first pump **3**L**1** is taken as an example and illustrated. FIG. **5**A is a cross-sectional view of a **5**A-**5**A cross-section of FIG. **2**. FIG. **5**B is a cross-sectional view of the first pump **3**L**1** illustrating a state in which the first eccentric shaft **5***e*L**1** has been rotated for 90 degrees from FIG. **5**A. FIG. **5**C is a cross-sectional view of the first pump **3**L**1** illustrating a state in which the first eccentric shaft **5***e*L**1** has been rotated for 90 degrees from FIG. **5**B. FIG. **5**D is a cross-sectional view of the first pump **3**L**1** illustrating a state in which the first eccentric shaft **5***e*L**1** has been rotated for 90 degrees from FIG. **5**C.

[0055] As illustrated in FIGS. **3** and **5**A, the first pump **3**L**1** includes a base member **36**L**1** fixed to the attachment member **37**, and a diaphragm member **31**L**1** nipped by the attachment member **37** and the base member **36**L**1**. The base member **36**L**1** includes an intake portion **34** serving as an air intake through which air is taken in, a discharge portion **35**L**1** serving as an air discharge port through which air is discharged, and a hole portion **36***a*L**1**. As illustrated in FIGS. **2**B and **5**A, the intake portion **34** is provided with an intake port not shown through which air is taken in, and a rib **34***a* that extends in the Z direction from the circumference of the intake port. Thereby, even if the member such as the third pipe **6**R**1** arranged in the vicinity of the intake portion **34** is sucked together with air toward the intake portion **34**, the member abuts against the rib **34***a*, such that the blocking of the intake portion **34** by the member may be suppressed. In the X direction orthogonal to the Y direction, one side thereof with respect to the rotational axis AL**1** is referred to as a first area AR**1**, and the other side thereof with respect to the rotational axis AL**1** is referred to as a second area AR**2**. That is, the first area AR**1** and the second area AR**2** may be disposed mutually on opposite sides of the rotational axis AL**1** in the X direction. In this state, the intake portion **34** is disposed in the first area AR**1**, and the discharge portion **35**L**1** is disposed in the second area AR**2**.

[0056] The diaphragm member **31**L**1** includes a flat surface portion **31***b*, and a pillar portion **31***b*L**1** that extends in the Z direction from the flat surface portion **31***b*. When the base member **36**L**1** and the diaphragm member **31**L**1** are attached to the attachment member **37**, the pillar portion **31***b*L**1** of the diaphragm member **31**L**1** passes through the hole portion **36***a*L**1** of the base member **36**L**1**. An upstream valve portion **31***c* and a downstream valve portion **31***d* are respectively disposed on an upstream end portion and a downstream end portion in the X direction of the flat surface portion **31***b*. A shaft hole portion **31***a*L**1** to which the first eccentric shaft **5***e*L**1** is inserted via a bearing **32** is disposed on the pillar portion **31***b*L**1**. In other words, the first eccentric shaft **5***e*L**1** is supported rotatably in the shaft hole portion **31***a*L**1** via the bearing **32**.

[0057] A recess portion **37***a* is disposed on the attachment member **37** at a position corresponding to the flat surface portion **31***b* of the diaphragm member **31**L**1**. In a state where the diaphragm member **31**L**1** and the base member **36**L**1** are attached to the attachment member **37**, a volume variation portion **33**L**1** is formed by the recess portion **37***a* of the attachment member **37** and the flat surface portion **31***b* of the diaphragm member **31**L**1**. The volume variation portion **33**L**1** is a space surrounded by the recess portion **37***a* and the flat surface portion **31***b*, and the first pump **3**L**1** is a diaphragm pump capable of varying the capacity of the volume variation portion **33**L**1**, which is the internal space of the first pump **3**L**1**.

[0058] As illustrated in FIG. **5**A, a state in which the flat surface portion **31***b* of the diaphragm member **31**L**1** is extended straight in the X direction is set as a normal state. By the first eccentric shaft **5***e*L**1** rotating about the first rotation shaft **4**Ls, the flat surface portion **31***b* of the diaphragm member **31**L**1** is displaced via the pillar portion **31***b*L**1**. As illustrated in FIG. **5**B, from a state in which the diaphragm member **31**L**1** is in a normal state, when the first eccentric shaft **5***e*L**1** is rotated for 90 degrees in the second rotation direction R**2**, the flat surface portion **31***b* of the diaphragm member **31**L**1** is displaced in a direction separating from the recess portion **37***a*, and the volume of the volume variation portion **33**L**1** is increased. In this state, the upstream valve portion **31***c* opens a path for air between the intake portion **34** and the volume variation portion **33**L**1**, and allows air to enter the volume variation portion **33**L**1**. Meanwhile, the downstream valve portion **31***d* shuts the path for air between the discharge portion **35**L**1** and the volume variation portion **33**L**1**, and regulates air from flowing in a reverse direction to the volume variation portion **33**L**1**.

[0059] As illustrated in FIG. **5**C, when the first eccentric shaft **5***e*L**1** is rotated for 90 degrees from the state illustrated in FIG. **5**B to the second rotation direction R**2**, the diaphragm member **31**L**1** returns to the normal state, and the volume of the volume variation portion **33**L**1** is reduced. Further, as illustrated in FIG. **5**D, when the first eccentric shaft **5***e*L**1** is rotated for 90 degrees from the state illustrated in FIG. **5**C to the second rotation direction R**2**, the flat surface portion **31***b* of the diaphragm member **31**L**1** is displaced to a direction approaching the recess portion **37***a*, and the volume of the volume variation portion **33**L**1** is reduced.

[0060] As described, in a state where the volume of the volume variation portion **33**L**1** is reduced, the upstream valve portion **31***c* shuts the path of air between the intake portion **34** and the volume variation portion **33**L**1**, and regulates air from entering the volume variation portion **33**L**1**. Meanwhile, the downstream valve portion **31***d* opens a path of air between the discharge portion **35**L**1** and the volume variation portion **33**L**1**, and allows discharge of air from the volume variation portion **33**L**1**. When the eccentric shaft **5***e*L**1** is rotated for 90 degrees in the second rotation direction R**2** from the state illustrated in FIG. **5**D, the diaphragm member **31**L**1** returns to the normal state and returns to the state illustrated in FIG. **5**A.

[0061] The diaphragm member **31**L**1** of the first pump **3**L**1** moves in reciprocating motion in the Z direction accompanying the rotation of the first eccentric shaft **5***e*L**1**, as illustrated in FIGS. **5**A to **5**D. Thereby, the first pump **3**L**1** functions as a diaphragm-type pump that takes in air through the intake portion **34** and discharges air through the discharge portion **35**L**1**.

[0062] The pump unit **1** includes the second pump **3**L**2**, the third pump **3**R**1**, and the fourth pump **3**R**2** that have a similar configuration as the first pump **3**L**1**. By the first rotation shaft **4**Ls of the first dual shaft motor **4**L rotating in a first rotation direction R**1**, drive is transmitted via the first drive transmission portion **5**L**1** to the first eccentric shaft **5***e*L**1**. In this state, by the second drive transmission portion **5**L**2** transiting to the non-transmission state, the rotation in the first rotation direction R**1** of the first rotation shaft **4**Ls is not transmitted to the second eccentric shaft **5***e*L**2**. Thereby, the first pump **3**L**1** will operate and the second pump **3**L**2** will not operate.

[0063] By the first rotation shaft **4**Ls of the first dual shaft motor **4**L rotating in the second rotation direction R**2**, drive is transmitted via the second drive transmission portion **5**L**2** to the second eccentric shaft **5***e*L**2**. In this state, by the first drive transmission portion **5**L**1** transiting to the non-transmission state, the rotation of the second rotation direction R**2** of the first rotation shaft **4**Ls is not transmitted to the first eccentric shaft **5***e*L**1**. Thereby, the second pump **3**L**2** will operate and the first pump **3**L**1** will not operate.

[0064] By the second rotation shaft **4**Rs of the second dual shaft motor **4**R rotating in the third rotation direction R**3**, drive is transmitted to the third eccentric shaft **5***e*R**1** via the third drive transmission portion **5**R**1**. In this state, by the fourth drive transmission portion **5**R**2** transiting to the non-transmission state, the rotation of the second rotation shaft **4**Rs in the third rotation direction R**3** is not transmitted to the fourth eccentric shaft **5***e*R**2**. Thereby, the third pump **3**R**1** will operate and the fourth pump **3**R**2** will not operate.

[0065] Further, by the second rotation shaft **4**Rs of the second dual shaft motor **4**R rotating in the fourth rotation direction R**4**, drive is transmitted to the fourth eccentric shaft **5***e*R**2** via the fourth drive transmission portion **5**R**2**. In this state, by the third drive transmission portion **5**R**1** transiting to the non-transmission state, the rotation of the second rotation shaft **4**Rs in the fourth rotation direction R**4** is not transmitted to the fourth eccentric shaft **5***e*R**2**. Thereby, the fourth pump **3**R**2** will operate and the third pump **3**R**1** will not operate.
[0066] As described, the pump unit **1** may cause arbitrary pumps (**3**L**1**, **3**L**2**, **3**R**1**, and **3**R**2**) to be operated by switching the rotation directions of the rotation shafts (**4**Ls and **4**Rs) of the two motors, which are the first dual shaft motor **4**L and the second dual shaft motor **4**R.
Pump Unit Layout
[0067] Next, a layout of the pump unit **1** will be described in detail. As illustrated in FIGS. **1** to **3**, the first rotation shaft **4**Ls of the first dual shaft motor **4**L and the second rotation shaft **4**Rs of the second dual shaft motor **4**R are arranged coaxially. That is, the first rotation shaft **4**Ls and the second rotation shaft **4**Rs are arranged on the same rotational axis AL**1**, as illustrated in FIG. **1**. Therefore, the size of the pump unit **1** in the X direction and the Z direction may be downsized.
[0068] The first dual shaft motor **4**L including the first housing **4**Lc, the first clutch portion **5***d*L**1**, the first eccentric shaft **5***e*L**1**, the second clutch portion **5***d*L**2**, and the second eccentric shaft **5***e*L**2** are aligned in the Y direction. Similarly, the second dual shaft motor **4**R including the second housing **4**Rc, the third clutch portion **5***d*R**1**, the third eccentric shaft **5***e*R**1**, the fourth clutch portion **5***d*R**2**, and the fourth eccentric shaft **5***e*R**2** are aligned in the Y direction. Therefore, the size of the pump unit **1** in the X direction and the Z direction may be downsized.
[0069] As illustrated in FIGS. **2**A to **2**C, the second discharge portion **35**L**2** and the third discharge portion **35**R**1** are arranged such that at least a portion thereof is mutually overlapped in the X direction, which is an orthogonal direction orthogonal to the Y direction. The second discharge portion **35**L**2** and the third discharge portion **35**R**1** are arranged between the two motors, which are the first dual shaft motor **4**L and the second dual shaft motor **4**R, in the Y direction. By adopting this arrangement, the size of the pump unit **1** in the Y direction may be downsized. The arrangements of the second discharge portion **35**L**2** and the third discharge portion **35**R**1** are not limited to the above-described arrangement.
[0070] Further, as illustrated in FIGS. **1** to **3**, the second pipe **6**L**2** and the third pipe **6**R**1** are respectively connected to the second discharge portion **35**L**2** and the third discharge portion **35**R**1**. The second pipe **6**L**2** is held by the second holding member **7**R holding the second housing **4**Rc of the second dual shaft motor **4**R, and the third pipe **6**R**1** is held by the first holding member **7**L holding the first housing **4**Lc of the first dual shaft motor **4**L.
[0071] As illustrated in FIG. **2**A, the second pipe **6**L**2** is arranged such that a portion **6**L**2***a* whose position in the Y direction overlaps with the second housing **4**Rc extends in parallel in the Y direction. The third pipe **6**R**1** is arranged such that a portion **6**R**1***a* whose position in the Y direction overlaps with the first housing **4**Lc extends in parallel in the Y direction. The portion **6**L**2***a* described above is held by hook portions **7**Ra and **7**Rb of the second holding member **7**R, as illustrated in FIGS. **2**A to **2**C and **3**, and the portion **6**R**1***a* is held by hook portions **7**La and **7**Lb of the first holding member **7**L. Further, the first discharge portion **35**L**1** is arranged on an opposite side as the second housing **4**Rc with respect to the first housing **4**Lc, and the first pipe **6**L**1** connected to the first discharge portion **35**L**1** extends to separate from the first housing **4**Lc in the Y direction. The fourth discharge portion **35**R**2** is arranged on an opposite side as the first housing **4**Lc with respect to the second housing **4**Rc, and the fourth pipe **6**R**2** connected to the fourth discharge portion **35**R**2** extends to separate from the second housing **4**Rc in the Y direction. By adopting such arrangement, the size of the pump unit **1** in the X direction and the Z direction may be downsized.
[0072] According further to the present embodiment, the second pipe **6**L**2**, the third pipe **6**R**1**, and the fourth pipe **6**R**2** are composed of a flexible member having flexibility against deformations

such as bends and expansion/contraction. Examples of a flexible member may include resin material such as plastic having flexibility, rubber, and cloth. The second pipe **6**L**2**, the third pipe **6**R**1**, and the fourth pipe **6**R**2** are formed in a tube shape, such that if the pipes are curved by an angle close to a right angle, the pipes may be bent and the paths in the interior of the pipe may be blocked. Therefore, the second pipe **6**L**2**, the third pipe **6**R**1**, and the fourth pipe **6**R**2** are configured to gradually change directions toward the downstream direction in the X direction, as illustrated in FIG. **1**.

[0073] Meanwhile, it is desirable that the first pipe **6**L**1** is bent by an angle close to a right angle, since space is limited. Therefore, the first pipe **6**L**1** is composed of a first member **6***a***1** that extends in the Y direction, a second member **6***a***2** that extends in the X direction, and a third member **6***a***3** that connects the first member **6***a***1** and the second member **6***a***2**, as illustrated in FIGS. **2**A, **6**A, and **6**B. The first member **6***a***1** is connected to the first discharge portion **35**L**1**. FIG. **6**A is a cross-sectional view of the first pipe **6**L**1**, and FIG. **6**B is a perspective view of the first pipe **6**L**1**. Further, FIG. **6**A is a **6**A-**6**A cross-section of FIG. **2**C.

[0074] The first member **6***a***1** and the second member **6***a***2** are each composed of a resin material having flexibility, and extend in mutually different directions. The third member **6***a***3** includes a bent portion **61** and has a higher stiffness than the stiffnesses of the first member **6***a***1** and the second member **6***a***2**. Further, the third member **6***a***3** includes a first connection portion **62** that extends in the Y direction and is connected to the first member **6***a***1**, and a second connection portion **63** that extends in the X direction and is connected to the second member **6***a***2**, wherein the bent portion **61** is formed between the first connection portion **62** and the second connection portion **63**.

[0075] By configuring the first pipe **6**L**1** as described above, the first pipe **6**L**1** may be bent by an angle close to a right angle without blocking the path on the inner side of the first pipe **6**L**1**. Further, the degree of freedom of arrangement of the first pipe **6**L**1** may be improved.

[0076] The first pipe **6**L**1**, the second pipe **6**L**2**, the third pipe **6**R**1**, and the fourth pipe **6**R**2** are not necessarily formed of flexible members, and may be composed of other materials, such as metal. Further, the second pipe **6**L**2**, the third pipe **6**R**1**, and the fourth pipe **6**R**2** may include a bent portion, similar to the first pipe **6**L**1**.

[0077] By configuring the pump unit **1** as described above, a new embodiment of a pump unit may be provided. Further, the pump unit **1** may be downsized.

Other Embodiments

[0078] According to the present embodiment, the first pump **3**L**1** is configured to be operated when the first rotation shaft **4**Ls is rotated in the first rotation direction R**1**, but the present technique is not limited thereto. Which pump is to be operated when the first rotation shaft **4**Ls or the second rotation shaft **4**Rs is rotated in the respective directions may be set arbitrarily.

[0079] According to the present embodiment, the first dual shaft motor **4**L, the first clutch portion **5***d*L**1**, the first eccentric shaft **5***e*L**1**, the second clutch portion **5***d*L**2**, and the second eccentric shaft **5***e*L**2** are arranged in an aligned manner in the Y direction, but the present technique is not limited thereto. For example, the first dual shaft motor **4**L, the first clutch portion **5***d*L**1**, the first eccentric shaft **5***e*L**1**, the second clutch portion **5***d*L**2**, and the second eccentric shaft **5***e*L**2** may be arranged in an aligned manner in either the X direction or the Z direction.

[0080] According to the present embodiment, the first rotation shaft **4**Ls of the first dual shaft motor **4**L and the second rotation shaft **4**Rs of the second dual shaft motor **4**R are arranged coaxially, but the present technique is not limited thereto. For example, the first rotation shaft **4**Ls and the second rotation shaft **4**Rs are not necessarily arranged coaxially, and may be somewhat displaced in the X direction or the Z direction

[0081] According to the present embodiment, the second pipe **6**L**2** is arranged such that the portion **6**L**2***a* whose position in the Y direction overlaps with the second housing **4**Rc extends in parallel in the Y direction. Further, the third pipe **6**R**1** is arranged such that the portion **6**R**1***a* whose position in

the Y direction overlaps with the first housing **4**Lc extends in parallel in the Y direction. However, the pump unit **1** is not limited to this arrangement, and the portions **6**L**2***a* and **6**R**1***a* are not necessarily arranged in parallel in the Y direction. Further, the first pipe **6**L**1** and the fourth pipe **6**R**2** may be arranged to respectively hold the first housing **4**Lc and the second housing **4**Rc. That is, the first pipe **6**L**1**, the second pipe **6**L**2**, the third pipe **6**R**1**, and the fourth pipe **6**R**2** are not necessarily arranged as described in the present embodiment, and arrangements thereof may be varied arbitrarily. The second pipe **6**L**2** and the third pipe **6**R**1** are each not necessarily held by the second housing **4**Rc and the first housing **4**Lc.
[0082] The respective pumps described above are not limited to the diaphragm-type pumps, and they may be composed of reciprocating pumps, such as piston pumps and plunger pumps.
[0083] While the present invention has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.
[0084] This application claims the benefit of Japanese Patent Application No. 2024-027891, filed Feb. 27, 2024, which is hereby incorporated by reference herein in its entirety.

Claims

1. A pump unit comprising: a motor including a rotation shaft configured to rotate about a rotational axis in a first rotation direction and in a second rotation direction opposite to the first rotation direction, the rotation shaft including a first end portion on one side of the rotation shaft and a second end portion on the other side of the rotation shaft in a direction of the rotational axis; a first pump and a second pump that are configured to send out air; a first drive transmission portion configured to be transitioned between a first transmission state in which a first driving force is transmitted from the first end portion of the rotation shaft to the first pump and a first non-transmission state in which the first driving force is not transmitted from the first end portion to the first pump; and a second drive transmission portion configured to be transitioned between a second transmission state in which a second driving force is transmitted from the second end portion of the rotation shaft to the second pump and a second non-transmission state in which the second driving force is not transmitted from the second end portion to the second pump, wherein the first drive transmission portion is configured to drive the first pump by being transitioned to the first transmission state when the rotation shaft rotates in the first rotation direction, and configured not to drive the first pump by being transitioned to the first non-transmission state when the rotation shaft rotates in the second rotation direction, and wherein the second drive transmission portion is configured not to drive the second pump by being transitioned to the second non-transmission state when the rotation shaft rotates in the first rotation direction, and configured to drive the second pump by being transitioned to the second transmission state when the rotation shaft rotates in the second rotation direction.
2. The pump unit according to claim 1, wherein the first drive transmission portion includes: a first clutch portion to which the first driving force from the first end portion is entered; and a first eccentric shaft connected to the first pump and arranged eccentrically with respect to the rotation shaft, wherein the second drive transmission portion includes: a second clutch portion to which the second driving force from the second end portion is entered; and a second eccentric shaft connected to the second pump and arranged eccentrically with respect to the rotation shaft, wherein the first clutch portion transmits the first driving force from the first end portion to the first eccentric shaft when the rotation shaft rotates in the first rotation direction, and the first clutch portion does not transmit the first driving force from the first end portion to the first eccentric shaft when the rotation shaft rotates in the second rotation direction, and wherein the second clutch portion does not transmit the second driving force from the second end portion to the second eccentric shaft

when the rotation shaft rotates in the first rotation direction, and the second clutch portion transmits the second driving force from the second end portion to the second eccentric shaft when the rotation shaft rotates in the second rotation direction.

3. The pump unit according to claim 2, wherein the motor, the first clutch portion, the first eccentric shaft, the second clutch portion, and the second eccentric shaft are aligned in the direction of the rotational axis.

4. The pump unit according to claim 1, wherein the first pump and the second pump are each a diaphragm pump.

5. The pump unit according to claim 1, wherein the rotation shaft and the motor are each a first rotation shaft and a first motor, wherein the pump unit further includes: a second motor including a second rotation shaft configured to rotate about the rotational axis in a third rotation direction and in a fourth rotation direction opposite to the third rotation direction, the second rotation shaft including a third end portion on one side of the second rotation shaft and a fourth end portion on the other side of the second rotation shaft in the direction of the rotational axis; a third pump and a fourth pump that are configured to send out air, a third drive transmission portion configured to be transitioned between a third transmission state in which a third driving force is transmitted from the third end portion of the second rotation shaft to the third pump and a third non-transmission state in which the third driving force is not transmitted from the third end portion to the third pump; and a fourth drive transmission portion configured to be transited between a fourth transmission state in which a fourth driving force is transmitted from the fourth end portion of the second rotation shaft to the fourth pump and a fourth non-transmission state in which the fourth driving force is not transmitted from the fourth end portion to the fourth pump, wherein the third drive transmission portion is configured to drive the third pump by being transitioned to the third transmission state when the second rotation shaft rotates in the third rotation direction, and configured not to drive the third pump by being transitioned to the third non-transmission state when the second rotation shaft rotates in the fourth rotation direction, and wherein the fourth drive transmission portion is configured not to drive the fourth pump by being transitioned to the fourth non-transmission state when the second rotation shaft rotates in the third rotation direction, and configured to drive the fourth pump by being transitioned to the fourth transmission state when the second rotation shaft rotates in the fourth rotation direction.

6. The pump unit according to claim 5, wherein the first pump, the second pump, the third pump and the fourth pump respectively include a first discharge portion, a second discharge portion, a third discharge portion and a fourth discharge portion, each of which discharges air, wherein the pump unit further includes: a first pipe connected to the first discharge portion and through which air discharged from the first discharge portion passes; a second pipe connected to the second discharge portion and through which air discharged from the second discharge portion passes; a third pipe connected to the third discharge portion and through which air discharged from the third discharge portion passes; and a fourth pipe connected to the fourth discharge portion and through which air discharged from the fourth discharge portion passes, wherein the first motor includes a first housing configured to support the first rotation shaft rotatably, wherein the second motor includes a second housing configured to support the second rotation shaft rotatably, wherein the second pipe is arranged such that a portion, of the second pipe, whose position in the direction of the rotational axis overlaps with the second housing extends in parallel with the direction of the rotational axis, and wherein the third pipe is arranged such that a portion, of the third pipe, whose position in the direction of the rotational axis overlaps with the first housing extends in parallel with the direction of the rotational axis.

7. The pump unit according to claim 6, further comprising: a first holding member configured to hold the first motor and the third pipe; and a second holding member configured to hold the second motor and the second pipe.

8. The pump unit according to claim 6, wherein the second discharge portion and the third discharge portion are arranged such that at least a portion of the second discharge portion and a portion of the third discharge portion are overlapped with each other when viewed in an orthogonal direction orthogonal to the direction of the rotational axis.

9. The pump unit according to claim 6, wherein the second discharge portion and the third discharge portion are arranged between the first housing and the second housing in the direction of the rotational axis.

10. The pump unit according to claim 6, wherein the first discharge portion is arranged on an opposite side from the second housing with respect to the first housing in the direction of the rotational axis, and wherein the fourth discharge portion is arranged on an opposite side from the first housing with respect to the second housing in the direction of the rotational axis.

11. The pump unit according to claim 6, wherein the first discharge portion, the second discharge portion, the third discharge portion and the fourth discharge portion extend in the direction of the rotational axis.

12. The pump unit according to claim 6, wherein at least one of the second pipe, the third pipe and the fourth pipe is composed of a resin material having flexibility.

13. The pump unit according to claim 6, wherein the first pipe includes: a first member and a second member that are each composed of a resin material having flexibility, and that extend in mutually different directions; and a third member including a bent portion and connecting the first member and the second member, and wherein a stiffness of the third member is higher than a stiffness of the first member and the second member.

14. The pump unit according to claim 5, wherein the first pump, the second pump, the third pump, and the fourth pump each include an intake portion through which air is taken in, and wherein the intake portion includes a rib arranged on a circumference of an intake port through which air is taken in.

15. An air supply device for conveying toner in an image forming apparatus, comprising: a motor including: (i) a housing; and (ii) first and second shaft portions which are rotatable together about a rotational axis in a first rotation direction and a second rotation direction opposite to the first rotation direction, and which extend in opposite directions with each other with respect to the housing in a direction of the rotational axis; first and second air supply unit configured to supply air; a first clutch configured to transmit a first driving force from the first shaft portion to the first air supply unit when the motor is rotated in the first rotation direction, and configured not to transmit the first driving force from the first shaft portion to the first air supply unit when the motor is rotated in the second rotation direction; and a second clutch configured to transmit a second driving force from the second shaft portion to the second air supply unit when the motor is rotated in the second rotation direction, and configured not to transmit the second driving force from the second shaft portion to the second air supply unit when the motor is rotated in the first rotation direction.

16. The air supply device according to claim 15, wherein the housing, the first clutch and the second clutch are aligned in the direction of the rotational axis.

17. The air supply device according to claim 15, wherein the housing and the first and second air supply units are aligned in the direction of the rotational axis.

18. The air supply device according to claim 15, wherein the first air supply unit includes a first air intake and a first air discharge port which are provided on a first area and a second area, respectively, the first area and the second area being provided on opposite sides with respect to the rotational axis in a direction orthogonal to the rotational axis.

19. The air supply device according to claim 15, wherein the motor, the housing and the rotational axis are a first motor, a first housing and a first rotational axis, respectively, and wherein the air supply device further includes: a second motor including: (i) a second housing; and (ii) third and fourth shaft portions which are rotatable together about a second rotational axis in a third rotation direction and a fourth rotation direction opposite to the third rotation direction, and which extend in opposite directions with each other with respect to the second housing in a direction of the second rotational axis; third and fourth air supply unit configured to supply air; a third clutch configured to transmit a third driving force from the third shaft portion to the third air supply unit when the second motor is rotated in the third rotation direction, and configured not to transmit the third driving force from the third shaft portion to the third air supply unit when the second motor is rotated in the fourth rotation direction; and a fourth clutch configured to transmit a fourth driving force from the fourth shaft portion to the fourth air supply unit when the motor is rotated in the fourth rotation direction, and configured not to transmit the fourth driving force from the fourth shaft portion to the fourth air supply unit when the motor is rotated in the third rotation direction.

20. An air supply device for conveying toner in an image forming apparatus, comprising: a motor including: (i) a housing; and (ii) first and second shaft portions which are rotatable together about a rotational axis in a first rotation direction and a second rotation direction opposite to the first rotation direction, and which extend in opposite directions with each other with respect to the housing in a direction of the rotational axis; first and second air supply unit configured to supply air, a first clutch configured to be transitioned between a first transmission state in which a first driving force is transmitted from the first shaft portion to the first air supply unit, and a first non-transmission state in which the first driving force is not transmitted from the first shaft portion to the first air supply unit; and a second clutch configured to be transitioned between a second transmission state in which a second driving force is transmitted from the second shaft portion to the second air supply unit, and a second non-transmission state in which the second driving force is not transmitted from the second shaft portion to the second air supply unit.